

Why 1.5 Is Used in the IQR Rule for Outlier Detection

- The interquartile (IQR) method of outlier detection uses 1.5 as its scale to detect outliers because it most closely follows Gaussian distribution. As a result, the method dictates that any data point that's 1.5 points below the lower bound quartile or above the upper bound quartile is an outlier.
- While working with a student of mine on an assignment applying the [interquartile range \(IQR\) method](#) of outlier detection, he asked me an intriguing question:
- "Why 1.5 times IQR? Why not one or two or any other number?"
- The IQR method of [outlier detection](#) is a method that dictates that any data point in a [boxplot](#) that's more than 1.5 IQR points below the first [quartile data](#) or more than 1.5 IQR points above the third quartile data is considered an outlier. Whether you're familiar with the rule or not, I hope the question makes you think about it. After all, isn't that what good data scientists do? Question everything, believe nothing.

Why Do You Multiply 1.5 in IQR Outlier Detection? Explained

- The interquartile (IQR) method of outlier detection uses 1.5 as its scale to detect outliers because it most closely follows Gaussian distribution. As a result, the method dictates that any data point that's 1.5 points below the lower bound quartile or above the upper bound quartile is an outlier.
- In the most general sense, an outlier is a data point which differs significantly from other observations. Now, its meaning can be interpreted according to the statistical model under study, but for the sake of simplicity and not to divert too far from the main purpose of this post, we'd consider first order statistics on a very simple data set, without any loss of generality.

What is the IQR Method of Outlier Detection?

- To explain IQR method easily, let's start with a [boxplot](#).
- A boxplot tells us, more or less, about the distribution of the data. It gives a sense of how much the data is actually spread out, what its range is and its skewness. A boxplot enables us to draw inferences from it for ordered data. It tells us about the various metrics of data arranged in ascending order.

In [this box plot](#), it labels the minimum on one side, maximum on the other and a median in the middle. That would mean:

- Minimum is the minimum value in the data set.
- Maximum is the maximum value in the data set.

So, the difference between the two tells us about the range of a data set.

- The [median](#) is the median (or center point), also called second quartile, of the data resulting from the fact that the data is ordered.
- $Q1$ is the first quartile of the data, which is to say 25 percent of the data lies between minimum and $Q1$.
- $Q3$ is the third quartile of the data, which is to say 75 percent of the data lies between minimum and $Q3$.
- The difference between $Q3$ and $Q1$ is called the interquartile range or IQR.
- $IQR = Q3 - Q1$
- To detect the outliers using this method, we define a new range, let's call it the decision range. Any data point lying outside this range is considered an outlier and is accordingly dealt with.
- The range is as given below:

Lower Bound: $(Q1 - 1.5 * IQR)$

Upper Bound: $(Q3 + 1.5 * IQR)$

Any data point less than the "Lower Bound" or more than the "Upper Bound" is considered an outlier.